Introduction to Economics

Class 7 — Elasticity

Pavel Kocourek

Universitat de Barcelona · Group A6

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Plan for today

Warm-up; Class 6, one more look
Olive oil abroad: a third less, the same money
Elasticity: the number, the midpoint
One line, many elasticities; in pairs
What makes demand elastic
Price $\times$ quantity: the box
Break
Topic fixed; Problem set 1 out
The olive oil answered; raise or cut?
Income and cross-price elasticities
Supply: when sellers can respond
In one line; before Monday 19 October

Warm-up: which curve moves, and which way?

Coach tickets, Barcelona–Valencia: for students, an inferior good

1. Students' incomes fall
2. Car-sharing rides get dearer
3. The coach fare falls
4. Diesel gets dearer

In pairs, two minutes; then hands per item.

Warm-up: what does the whole market buy?

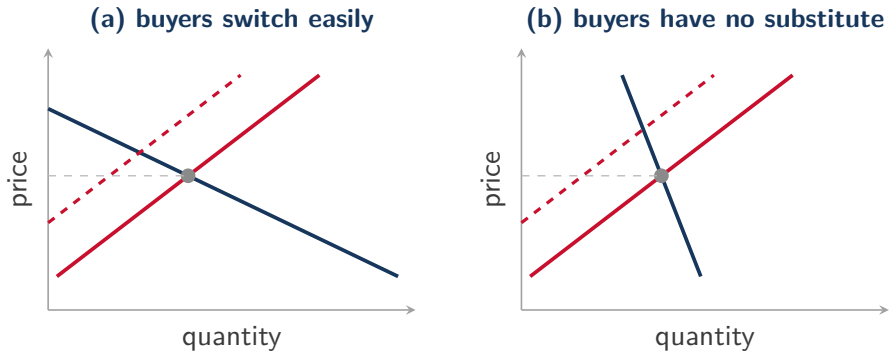
Padel-court hours a month

price	Ana	Biel	the market
10 euros	5	3	
20 euros	3	1	

Carla joins: 2 hours at 10 euros, 1 at 20. The market curve?

In pairs, one minute; then hands.

Class 6 again: the same shift — where does the price jump more?



Olive oil abroad in 2023: a third less oil, the same money

1.07 million tonnes → **0.68 million tonnes**

shipped abroad, 2022 → 2023

4.10 bn € → **4.15 bn €**

what came back, 2022 → 2023

What does that say about the buyers abroad?

Spain's olive-oil exports, customs data (AEAT), in MAPA's 2023 product report; 2023 provisional.

Elasticity: how much the quantity moves, in percentages

$$\text{price elasticity of demand} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}}$$

- > 1 **elastic**: the quantity moves more than the price
- < 1 **inelastic**: the quantity moves less than the price
- $= 1$ **unit elastic**: the same

Percentages, so cups and tonnes compare. The minus sign is dropped.

Two points, one number: percentages of the midpoint

The neighbourhood's cups: 2.00 → 2.50, 160 → 120 a day

computed from	price	cups	elasticity
the start, 2.00	+25 %	−25 %	1.0
the start, 2.50	−20 %	+33 %	1.67

Two points, one number: percentages of the midpoint

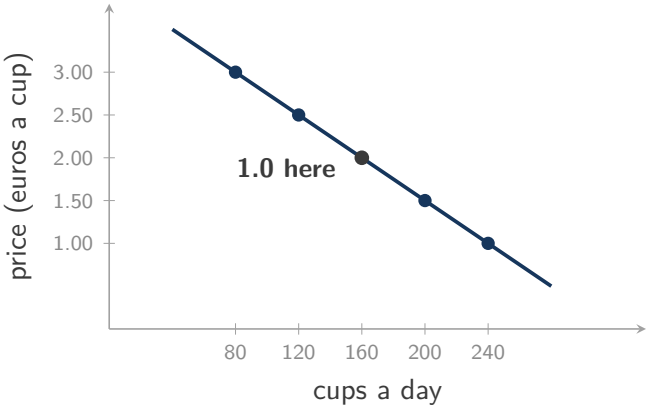
The neighbourhood's cups: 2.00 → 2.50, 160 → 120 a day

computed from	price	cups	elasticity
the start, 2.00	+25 %	−25 %	1.0
the start, 2.50	−20 %	+33 %	1.67
the midpoint	**+22 %**	**−29 %**	**1.29**

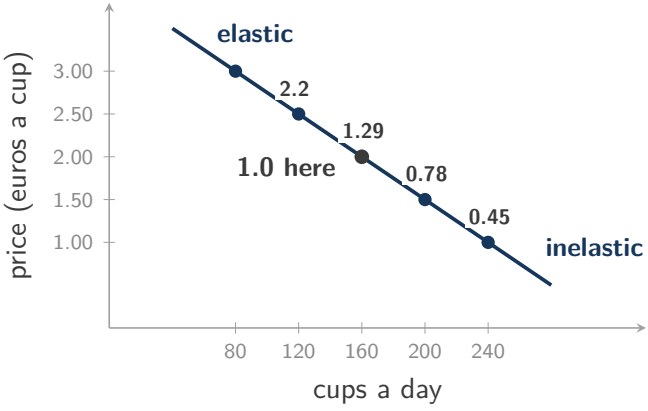
$$\frac{(Q_2 - Q_1) \,/\, \tfrac{1}{2}(Q_1 + Q_2)}{(P_2 - P_1) \,/\, \tfrac{1}{2}(P_1 + P_2)}$$

Divide each change by the average of the two values, not by the start.

One straight line: elastic at the top, inelastic at the bottom



One straight line: elastic at the top, inelastic at the bottom



Same slope everywhere; the percentages change:
few cups and a high price at the top, many cups and a low price at the bottom.

In pairs, four minutes: compute, classify, and one you cannot

Elastic, inelastic — or no number?

1. Bikinis at the bars: 5.00 → 5.50 euros, 200 → 150 a day

2. A single metro fare +10 %; rush-hour journeys −2 %

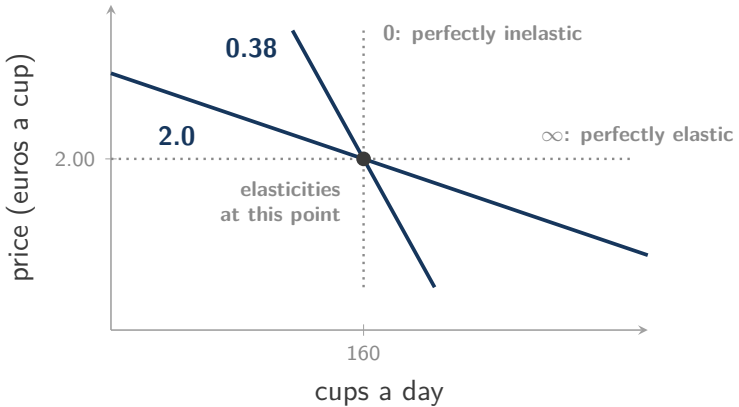
3. The T-usual halved in Sep 2022; all journeys in 2023 +17 %

Midpoint method where you have two points; then hands per item.

Demand is elastic when buyers can walk away

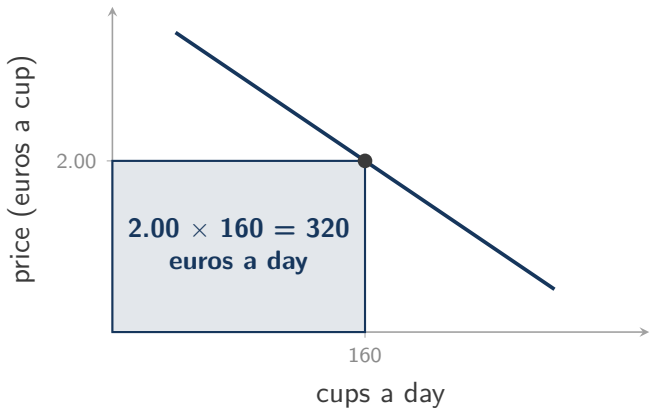
- Close substitutes nearby: Bar Nou, or the bar next door
- A luxury, not a necessity: a dinner out, not bread
- A narrow market: one bar's cup, not "coffee"
- Time to adjust: a year, not a week
- A big slice of the budget: rent, not chewing gum

Through one point: the flatter the line, the more elastic



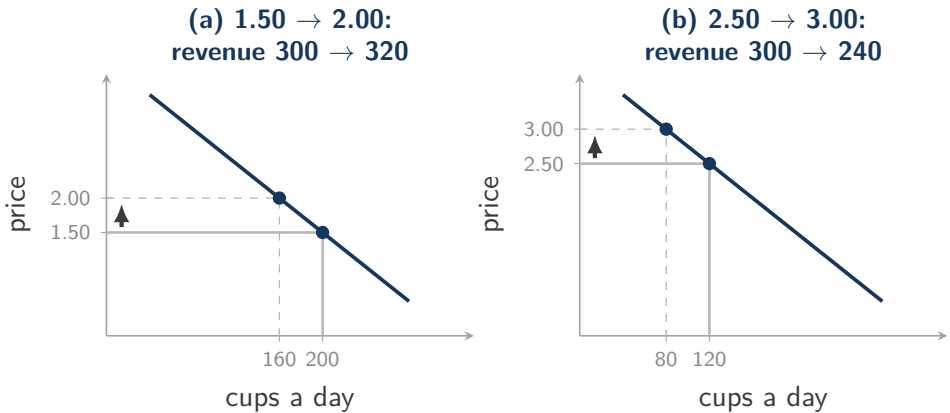
Same axes only. A line that looks flat on a stretched axis means nothing.

What buyers spend is what sellers get: price × quantity, a box

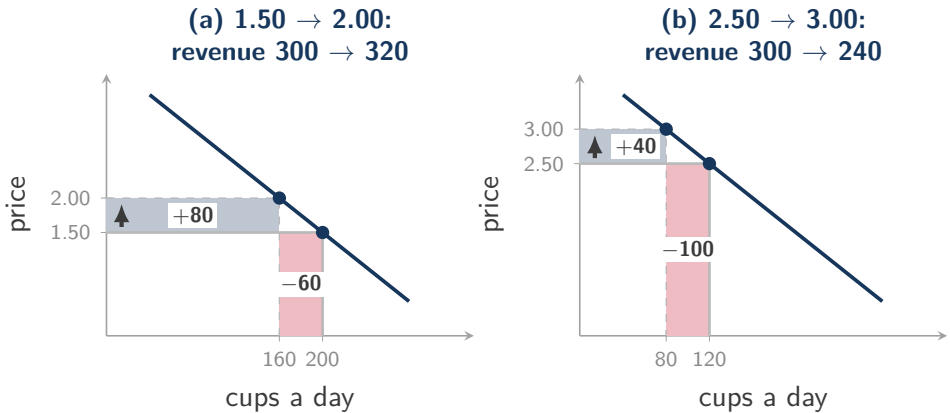


Total expenditure of buyers = **total revenue** of sellers: one box, two names.

A dearer cup: more money at the bottom of the line, less at the top



A dearer cup: more money at the bottom of the line, less at the top



Inelastic stretch (0.78): the gained strip wins. Elastic stretch (2.2): the lost one does.

Raise the price when demand is inelastic; cut it when elastic

demand	price up	price down
inelastic (< 1)	revenue up	revenue down
unit ($= 1$)	unchanged	unchanged
elastic (> 1)	revenue down	revenue up

Inelastic: revenue follows the price. Elastic: it runs the other way.

Which curve moves, and how many members? (optional)

Gym memberships in a city; price elasticity of demand 1.5

1. Memberships get 10 % dearer
2. Workout apps get dearer
3. Gyms' electricity bills double

In pairs, two minutes; then hands per item.

Break

Ten minutes

Today: your topic is fixed; Problem set 1 is out

- Project: each group's question is settled today
- Not settled? See me after class
- Problem set 1 (Blocks 1–2): out today
- Due Thursday 22 October

Revealed in class

Before class: slide 6's two pairs of numbers, the midpoint method — what elasticity do they give, and what does it say about the money?

In pairs: raise the price, or cut it?

More money, or less?

1. Rush-hour metro, elasticity 0.2: a dearer fare
2. Cinema on a Tuesday afternoon, 2.5: a cheaper ticket
3. Bar Nou alone among twelve bars, 4.0: a dearer cup
4. The neighbourhood's cups: 2.00 $\rightarrow$ 2.50, 160 $\rightarrow$ 120 a day

Three minutes; then hands per item.

Income elasticity: more income, more of most goods, less of some

$$\text{income elasticity of demand} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in income}}$$

- **Normal good**: positive — a dinner out, a flat white
- **Inferior good**: negative — instant noodles, the cheapest own brand
- Among normal goods: luxuries above 1, necessities below 1

Class 5's 'income' shifter, now with a sign and a size.

Cross-price elasticity: substitutes positive, complements negative

$$\text{cross-price elasticity} = \frac{\%\text{ change in quantity of A}}{\%\text{ change in price of B}}$$

- The T-usual half price $\rightarrow$ fewer T-casual journeys: **substitutes**, positive
- Croissants dearer $\rightarrow$ fewer cafès amb llet: **complements**, negative

Zero: unrelated goods. Class 5's shift of demand, measured.

In pairs: which elasticity, what sign, what word?

Name it, compute it, give the word.

1. Income $+10\,\%$; instant noodles $-5\,\%$
2. Income $+10\,\%$; concert tickets $+15\,\%$
3. Tea $+20\,\%$; cups of coffee $+4\,\%$
4. Croissants $+10\,\%$; cafès amb llet $-3\,\%$
5. A cup $+10\,\%$; cups $-8\,\%$

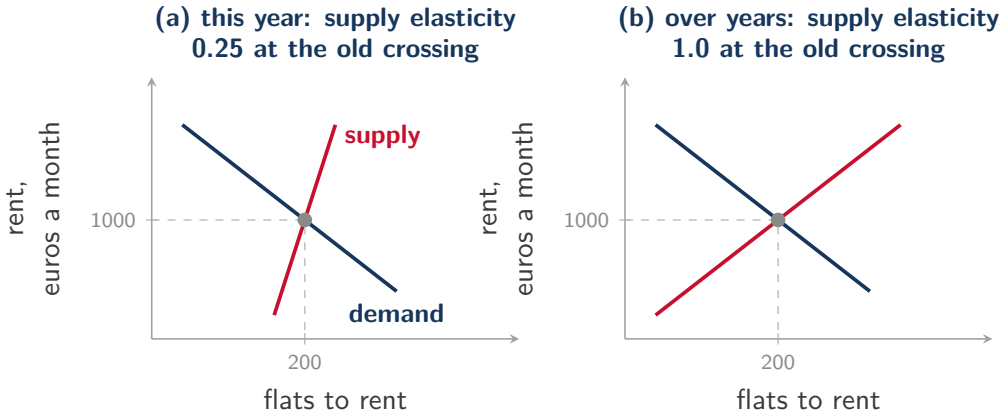
Three minutes; then hands per item.

Supply elasticity: sellers respond when they can; time is the key

$$\text{price elasticity of supply} = \frac{\%\text{ change in quantity supplied}}{\%\text{ change in price}}$$

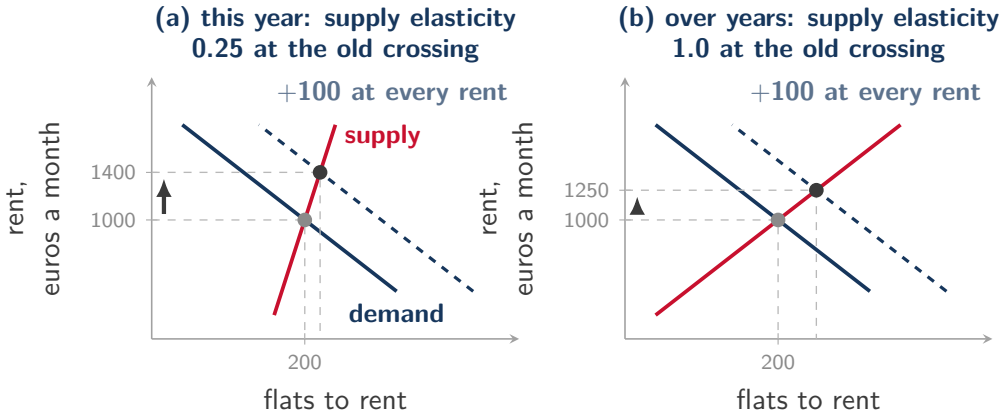
- This year's olives are already on the trees: near 0
- Spare capacity, stock in the warehouse: elastic
- Years to plant, build, hire: more elastic
- Vertical: perfectly inelastic — the seats in a hall tonight

Flats in Gràcia: more people want one; this year the rent jumps



Own numbers; not data.

Flats in Gràcia: more people want one; this year the rent jumps



Same shift. Steep supply: rent +400, flats +20. Flatter: rent +250, flats +50.

Own numbers; not data.

Revealed in class

Before class: slide 5 once more — in which panel does the price jump more, and in which does the quantity fall more? Say why, given that the shift is the same.

Two juice stalls: who should make what? (optional)

In one hour: Red stall, 6 smoothies or 3 sandwiches; Blue stall, 2 or 2

1. Red is faster at both. Can trade still help?
2. Who makes the sandwiches?
3. A sandwich price both accept?
4. Blue's hour becomes 2 smoothies or 1 sandwich. Now?

What moves on the frontier? (optional)

A region grows almonds and makes bicycles

1. A better almond variety doubles the harvest
2. Laid-off bicycle workers get their jobs back
3. 5,000 workers move in, able to do either job

In pairs; draw it first.

In one line

Revealed in class

Before class: try to say the whole class in two lines — what elasticity measures, and what it decides when supply shifts.

Before Monday 19 October

- Block 3 begins: surplus, efficiency, taxes and subsidies
- Read Mankiw & Taylor, Ch. 6, pp. 141–158 — consumer and producer surplus, market efficiency; Ch. 7, pp. 163–171 — taxes and subsidies
- Problem set 1 (Blocks 1–2): due Thursday 22 October
- Practise in Sextant: the units the course page lists

pkocourek@ub.edu · pkocourek.com/teaching/intro-economics · pkocourek.com/sextant/?c=intro-econ